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```
clc; clear; close all;
```

IIR Filter Coefficients

```
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

N = 50;
x = [1 zeros(1,N-1)];       % Unit impulse
n = 0:N-1;
```

1. Direct Form-I and Direct Form-II

```
y1 = filter(b,a,x);         % Direct Form-I
y2 = filter(b,a,x);         % Direct Form-II
```

2. Cascade Realization

```
[sos,g] = tf2sos(b,a);
yc = g*x;

for k = 1:size(sos,1)
    yc = filter(sos(k,1:3),sos(k,4:6),yc);
end
```

3. Parallel Realization

```
[r,p,k] = residuez(b,a);
yp = filter(k,1,x);

for i = 1:length(r)
    yp = yp + filter(r(i),[1 -p(i)],x);
end
```

4. Display Coefficients

```
disp('Numerator coefficients b ='); disp(b)
disp('Denominator coefficients a ='); disp(a)

disp('Second Order Sections (Cascade) ='); disp(sos)
disp('Cascade Gain ='); disp(g)

disp('Residues (Parallel) ='); disp(r)
disp('Poles (Parallel) ='); disp(p)
disp('Direct term ='); disp(k)
```

```
Numerator coefficients b =
    0.0048    0.0193    0.0289    0.0193    0.0048

Denominator coefficients a =
    1.0000   -2.3695    2.3140   -1.0547    0.1874

Second Order Sections (Cascade) =
    1.0000    2.1551    1.0000    1.0000   -1.0486    0.2962
    1.0000    1.8657    1.0000    1.0000   -1.3209    0.6327

Cascade Gain =
    0.0048

Residues (Parallel) =
  -0.2698 + 0.1343i
  -0.2698 - 0.1343i
   0.2594 - 0.8024i
   0.2594 + 0.8024i

Poles (Parallel) =
   0.6604 + 0.4433i
   0.6604 - 0.4433i
   0.5243 + 0.1459i
   0.5243 - 0.1459i

Direct term =
    0.0256
```

5. Error and Percentage Deviation

```
E = [max(abs(y1-y2))
    max(abs(y1-yc))
    max(abs(y1-yp))];

D = E/max(abs(y1))*100;

T = table(["DF-I";"DF-II";"Cascade";"Parallel"], ...
    [0;E], ...
    [0;D], ...
    'VariableNames',{'Structure','Maximum_Error','Deviation_percent'});
```

```
disp('Comparison Table:')
disp(T)
```

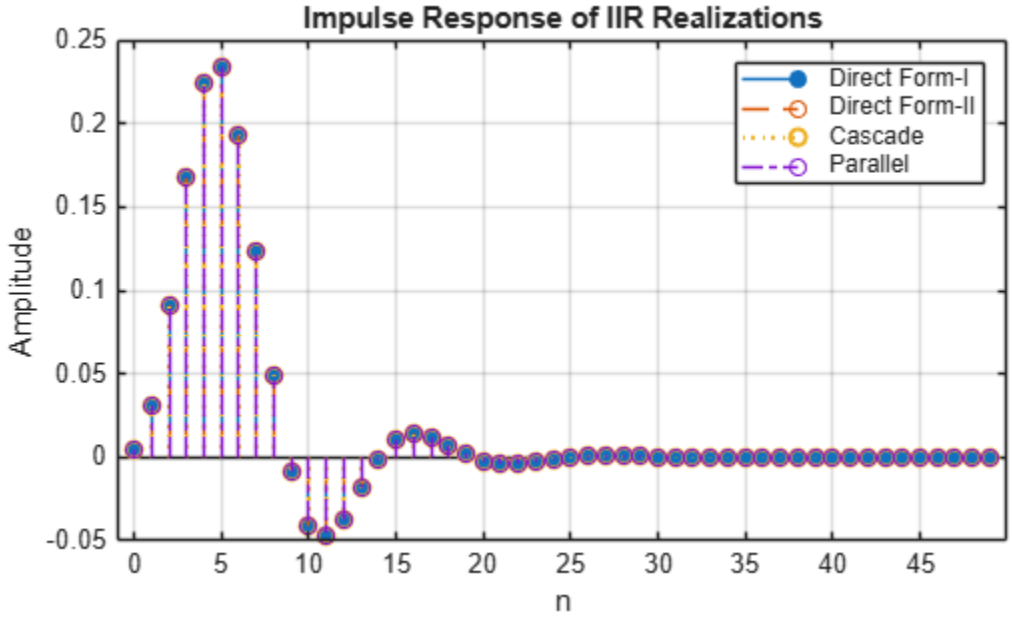
```
Comparison Table:
    Structure     Maximum_Error    Deviation_percent
    __________    _____________    _________________

    "DF-I"                   0                    0
    "DF-II"                  0                    0
    "Cascade"       1.9221e-15           8.242e-13
    "Parallel"      1.9811e-15          8.4949e-13
```

6. Impulse Response Comparison

```
figure;
stem(n,y1,'filled'); hold on;
stem(n,y2,'--');
stem(n,yc,':','LineWidth',1.5);
stem(n,yp,'-.');
grid on;
xlabel('n');
ylabel('Amplitude');
title('Impulse Response of IIR Realizations');
legend('Direct Form-I','Direct Form-II','Cascade','Parallel');
```

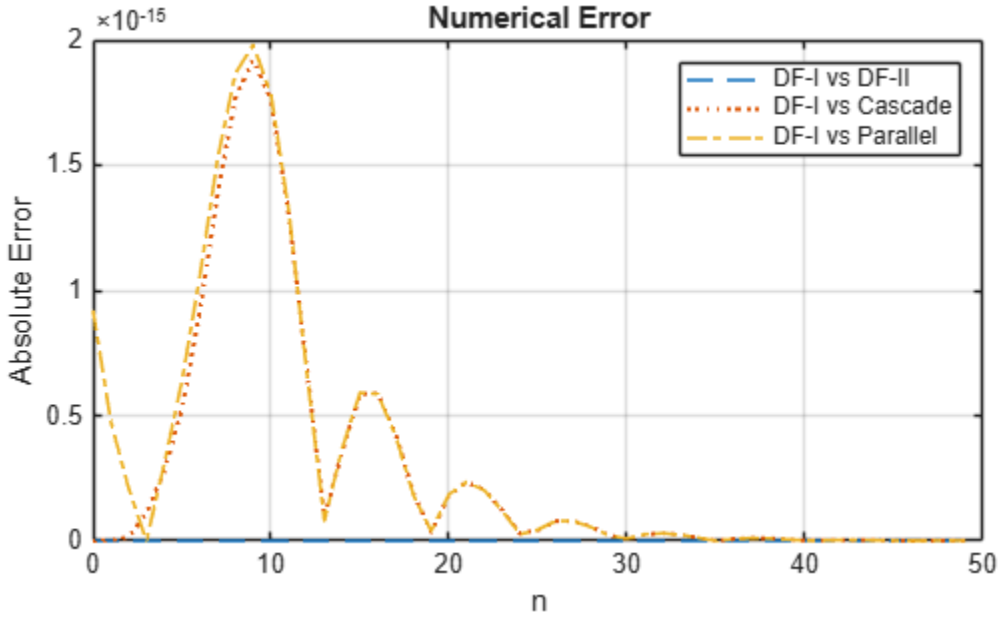
```
Warning: Using only the real component of complex data.
```



7. Error Plot

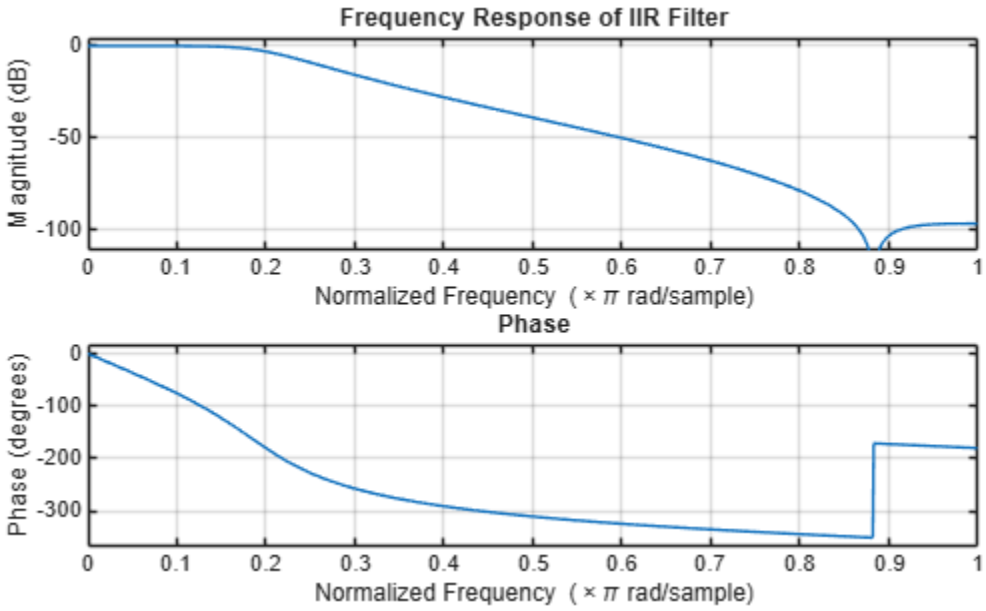
```
figure;
plot(n,abs(y1-y2),'--','LineWidth',1.2); hold on;
```

```
plot(n,abs(y1-yc),':','LineWidth',1.5);
plot(n,abs(y1-yp),'-.','LineWidth',1.2);
grid on;
xlabel('n');
ylabel('Absolute Error');
title('Numerical Error');
legend('DF-I vs DF-II','DF-I vs Cascade','DF-I vs Parallel');
```



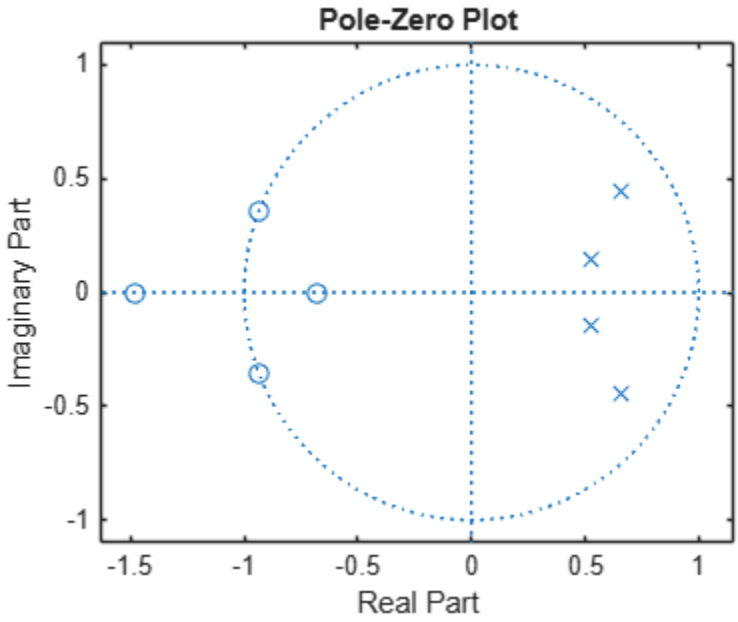
8. Frequency Response

```
figure;
freqz(b,a,1024);
title('Frequency Response of IIR Filter');
```



9. Pole-Zero Plot

```
figure;
zplane(b,a);
title('Pole-Zero Plot');
```



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